

CHI-SQUARE TEST OF INDEPENDENCE

Concept, Calculation, Interpretation & Chi-Square-Based Visualization

A Data Analyst + Statistics Study Guide
Prepared for: B.Tech Data Analytics Students

NOTE ON SOURCE MATERIAL

A note on sources: This document was requested with the uploaded course material as the primary source for definitions, terminology, formulas, and examples. No course-material file (PDF, slide, or image) was actually attached to the request — only the content brief was received. To keep this useful, the notes below follow the standard chi-square-test conventions used across virtually all data analytics courses (Pearson's chi-square statistic, O_{ij}/E_{ij} notation, the critical-value and p-value decision rules, and Pearson residuals) and uses the exact Gender $\times$ Admission Status example given in the brief. If the real course file is shared later, this document can be re-aligned to match its exact wording and numbers.

1. INTRODUCTION

Every Data Analyst eventually runs into a question that looks simple but needs a formal test to answer properly: “Are these two categorical variables associated with each other?” For example: does a customer's segment relate to which product they prefer? Does gender relate to admission outcome? The Chi-Square Test of Independence is the standard statistical tool for answering exactly this kind of question, using nothing more than counts of categories.

Key terms, in plain language first

- Attribute / categorical variable — a variable whose values are categories or labels, not numbers you can average — e.g. Gender (Male/Female), Admission Status (Yes/No), Customer Segment (Premium/Standard).
- Independence — two variables are independent if knowing the value of one gives you no extra information about the other. If Gender and Admission Status are independent, the admission rate should look the same for men and women.
- Dependence — the opposite: knowing one variable changes what you'd expect for the other. If admission rates clearly differ by gender in the data, that's dependence.
- Association — the general term for “a relationship exists between the two variables” — used interchangeably with dependence in this context.
- Contingency table — the table of counts that cross-tabulates the two variables. We need it because the entire test is built from counts, not from raw individual observations — see Section 2.

What is the Chi-Square Test of Independence trying to detect?

It is trying to detect whether the pattern of counts actually observed in a sample is close enough to the pattern we'd expect under independence — or far enough away from it that random sampling alone is an unconvincing explanation. It does this by comparing observed frequencies against expected frequencies under independence, and summarizing the total mismatch in a single number, the chi-square (χ^2) statistic.

Working example used throughout this document

Gender × Admission Status. Question: “Is admission status independent of gender, or is there evidence of an association?” This exact example (with the contingency table below) is carried through every section of this document — the same numbers are used for every calculation, every chart, and every code example, so nothing is recomputed from a different dataset partway through.

2. CONTINGENCY TABLE

A contingency table (also called a cross-tabulation) lays out the joint counts of two categorical variables — one variable's categories along the rows, the other's along the columns. Here is the working example:

	Admission: Yes	Admission: No	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

Reading the table

- Rows — the categories of the first variable: Male, Female.
- Columns — the categories of the second variable: Admission Yes, Admission No.
- Row totals — total count for each row, ignoring column: Male = 50, Female = 50.
- Column totals — total count for each column, ignoring row: Yes = 40, No = 60.
- Grand total — total number of observations in the whole table: 100.
- Individual cell frequency — the count for one specific row-column combination, e.g. Male & Yes = 30.

Notation

O_{ij} = Observed frequency in cell (row i , column j)

For instance O_{11} is the observed count in row 1, column 1 (Male, Yes) = 30.

The values inside the table are called observed frequencies because they are exactly what actually happened in the collected sample — real counts, not predictions. This table is the starting point for the entire chi-square procedure: every later formula (expected frequency, the χ^2 statistic, residuals) is computed directly from the row totals, column totals, and grand total shown here.

3. OBSERVED FREQUENCY

Observed frequency is simply the actual frequency found in the data — no calculation involved, just a count from the contingency table.

Cell	Meaning	Observed Value
O ₁₁	Male & Admission = Yes	30
O ₁₂	Male & Admission = No	20
O ₂₁	Female & Admission = Yes	10
O ₂₂	Female & Admission = No	40

That's the whole idea — O₁₁ = 30 simply means “30 people in this dataset are Male and were Admitted.” Nothing more complicated than reading a number off the table.

4. EXPECTED FREQUENCY

This is the key idea the whole test is built on. Expected frequency means: “the frequency we would expect to see in a given cell if the two categorical variables were independent.” It is not a real observed count — it's a hypothetical benchmark computed purely from the row and column totals.

Expected Frequency Formula

$$E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$$

Calculating every expected frequency

Male + Yes:

$$E = (50 \times 40) / 100 = 20$$

Male + No:

$$E = (50 \times 60) / 100 = 30$$

Female + Yes:

$$E = (50 \times 40) / 100 = 20$$

Female + No:

$$E = (50 \times 60) / 100 = 30$$

Complete Expected-Frequency Table

	Admission: Yes	Admission: No
Male	20	30
Female	20	30

OBSERVED vs EXPECTED — the core distinction

Observed = what actually happened in the sample.

Expected = what would happen, approximately, if independence were true — it is a reference point for comparison, never a target or a “correct” value.

5. OBSERVED VS EXPECTED

OBSERVED

	Yes	No
Male	30	20
Female	10	40

EXPECTED

	Yes	No
Male	20	30
Female	20	30

Male + Yes: Observed = 30, Expected = 20 → Difference = +10

Female + Yes: Observed = 10, Expected = 20 → Difference = -10

THE CENTRAL IDEA OF THIS ENTIRE TEST

The Chi-square test of independence is fundamentally based on comparing OBSERVED FREQUENCIES with EXPECTED FREQUENCIES. Every remaining step in this document — the χ^2 statistic, the p-value, residuals, and every visualization — is just a different way of measuring, decomposing, or displaying this one O-vs-E comparison.

6. CHI-SQUARE STATISTIC

Chi-Square Statistic

$$\chi^2 = \sum [(O_{ij} - E_{ij})^2 / E_{ij}]$$

- O_{ij} = Observed frequency in the cell.
- E_{ij} = Expected frequency in the cell.
- $O_{ij} - E_{ij}$ = the raw difference between what happened and what independence predicts.
- $(O_{ij} - E_{ij})^2 / E_{ij}$ = the contribution of that single cell to the overall χ^2 statistic.

Calculation table

Cell	Observed (O)	Expected (E)	O - E	(O-E) ² / E
Male, Yes	30	20	+10	5.000
Male, No	20	30	-10	3.333
Female, Yes	10	20	-10	5.000
Female, No	40	30	+10	3.333
TOTAL				$\chi^2 = 16.667$

Why does a bigger gap give a bigger χ^2 ?

If O and E are very similar → small difference → small contribution → small χ^2 .

If O and E are very different → large difference → larger contribution → larger χ^2 . Squaring the difference means bigger mismatches are penalized disproportionately more.

7. CHI-SQUARE CONTRIBUTION

Individual cell contributions deserve special attention — they are what a Data Analyst actually inspects after getting a significant result.

$$\text{Cell Contribution} = (O_{ij} - E_{ij})^2 / E_{ij}$$

Cell Contributions Sum to χ^2

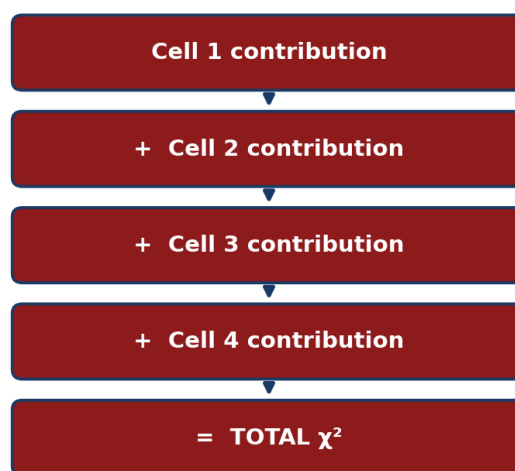


Figure: The total χ^2 statistic is nothing more than the sum of every cell's own contribution.

Why this matters for a Data Analyst

The overall χ^2 tells you whether the table shows evidence of association at all.

The individual contributions tell you which specific cells are driving that statistic — this is what turns a yes/no test result into an actionable business insight (see Section 14 and Section 22).

8. HYPOTHESES

H_0 — Null Hypothesis
The two attributes are independent.

H_1 — Alternative Hypothesis
The two attributes are dependent.

In the Gender $\times$ Admission example: H_0 states that Gender and Admission Status are independent — admission rates should not differ meaningfully between Male and Female applicants. H_1 states the opposite — that admission status does depend on gender in this population. The chi-square test uses the sample data to decide which of these two statements the evidence supports.

9. DEGREES OF FREEDOM

Degrees of Freedom
 $df = (m - 1)(n - 1)$

m = number of rows, n = number of columns

Table size	Calculation	df
2×2	$(2-1)(2-1)$	1
3×3	$(3-1)(3-1)$	4
3×4	$(3-1)(4-1)$	6
4×4	$(4-1)(4-1)$	9

For the working example (2×2 table): $df = 1$.

EXAM / ANALYSIS TRAP

Do not calculate degrees of freedom as $m \times n$ (total number of cells). Correct: $df = (m-1)(n-1)$.

10. ASSUMPTIONS

The chi-square test of independence relies on a small set of assumptions that should be checked before trusting the p-value:

- Sample size should be sufficiently large. The chi-square distribution is used as an approximation to the true sampling distribution of the test statistic, and that approximation only holds well when the sample is large enough.
- Cell frequencies should satisfy the required expected-frequency condition. As a standard requirement, expected cell counts should not be too small (the commonly used convention is an expected count of at least 5 in the large majority of cells) — use the specific threshold given in your own course material if it differs from this convention.
- Observations should be independent of one another — each unit in the sample should be counted in exactly one cell, and one observation's category should not influence another's.

Why assumptions matter

If these assumptions are violated — e.g., the sample is small or several expected counts are very low — the calculated p-value can be unreliable, and the resulting decision (reject / do not reject H_0) may not be trustworthy. Always check assumptions before interpreting the result, not after.

11. DECISION USING CRITICAL VALUE

Critical-Value Decision Rule

If $\chi^2(\text{calculated}) > \chi^2(\text{critical}) \rightarrow \text{Reject } H_0$
If $\chi^2(\text{calculated}) \leq \chi^2(\text{critical}) \rightarrow \text{Do not reject } H_0$

Critical-Value Decision Flowchart

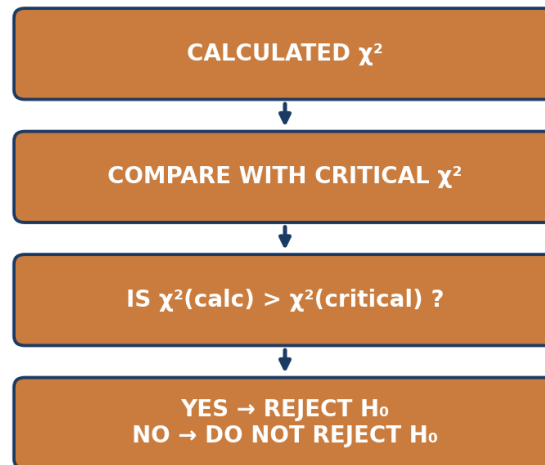


Figure: Critical-value decision flowchart.

For the working example, $df = 1$. The critical value at $\alpha = 0.05$ is 3.841, and at $\alpha = 0.01$ is 6.635. Since the calculated $\chi^2 = 16.667$ is greater than both critical values, H_0 is rejected at both significance levels.

12. DECISION USING P-VALUE

P-value Decision Rule

$p\text{-value} < \alpha \rightarrow \text{Reject } H_0$
 $p\text{-value} \geq \alpha \rightarrow \text{Do not reject } H_0$

For the working example: $p\text{-value} \approx 0.0000446$, which is far below both $\alpha = 0.05$ and $\alpha = 0.01$ — so H_0 is rejected.

WORDING THAT MUST BE AVOIDED

Do NOT say: “the p-value is the probability that H_0 is true.” This is a common and serious misinterpretation — the p-value is calculated assuming H_0 is true, so it cannot simultaneously be “the probability that H_0 is true.”

Do NOT say: “failing to reject H_0 proves independence.” Failing to reject only means the data did not give strong enough evidence against independence — not that independence has been proven.

Statistically appropriate wording

Instead of claiming proof, say: “There is insufficient evidence to conclude that the variables are associated.” This is the correct, defensible way to report a non-significant result.

13. ASSOCIATION IS NOT CAUSATION

ASSOCIATION $\neq$ CAUSATION

The chi-square test can provide evidence of an association between two categorical variables. It does NOT establish that one variable causes changes in the other.

Simple example: suppose the test shows a significant association between Gender and Admission Status. This tells us admission outcomes differ by gender in this sample — it does not tell us gender causes that difference. Some other factor (for example, which departments each group tends to apply to — see the Berkeley Admissions example in Section 18) could be driving the pattern instead.

A Data Analyst should report the result strictly as an observed statistical association, e.g., “The Chi-square test provides evidence of an association between Gender and Admission Status” — never as a causal claim such as “Gender causes different admission outcomes.”

14. THE CORE DATA ANALYST QUESTION

A complete chi-square analysis really answers three separate, layered questions:

#	Question	Answered by
1	Is there evidence of an association?	CHI-SQUARE TEST
2	Which cells are responsible for the deviation?	RESIDUAL / CELL CONTRIBUTION ANALYSIS
3	How can I communicate those deviations visually?	CHI-SQUARE-RELATED VISUALIZATION

From Table to Insight — The Core Data Analyst Path

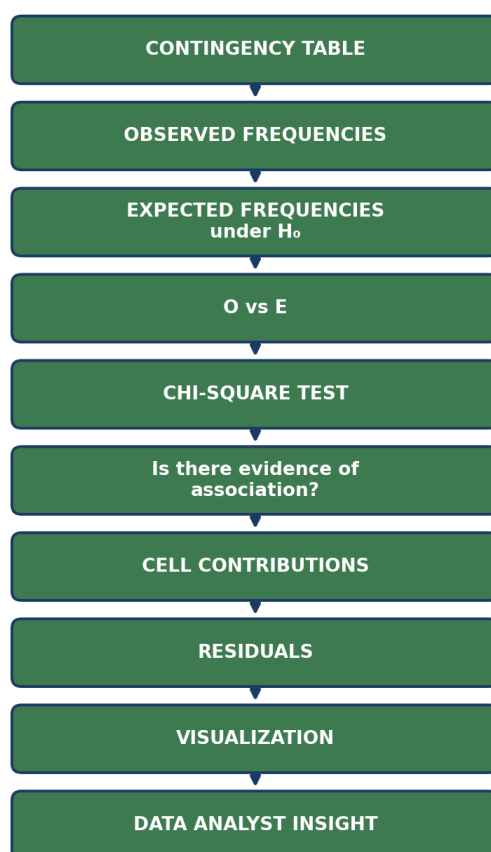


Figure: the path from a contingency table to a communicable insight.

15. CHI-SQUARE TEST VISUALIZATION

SCOPE OF THIS SECTION

Every visualization in this section is directly connected to the chi-square test itself — observed frequencies, expected frequencies, the O-E difference, chi-square contributions, or residuals. Generic categorical charts that don't connect back to these quantities are intentionally left out of this section.

15.1 Observed vs Expected

The most direct visualization compares observed and expected frequencies side by side, using the exact same contingency table used in the chi-square calculation above.

Observed vs Expected Frequencies — Gender × Admission

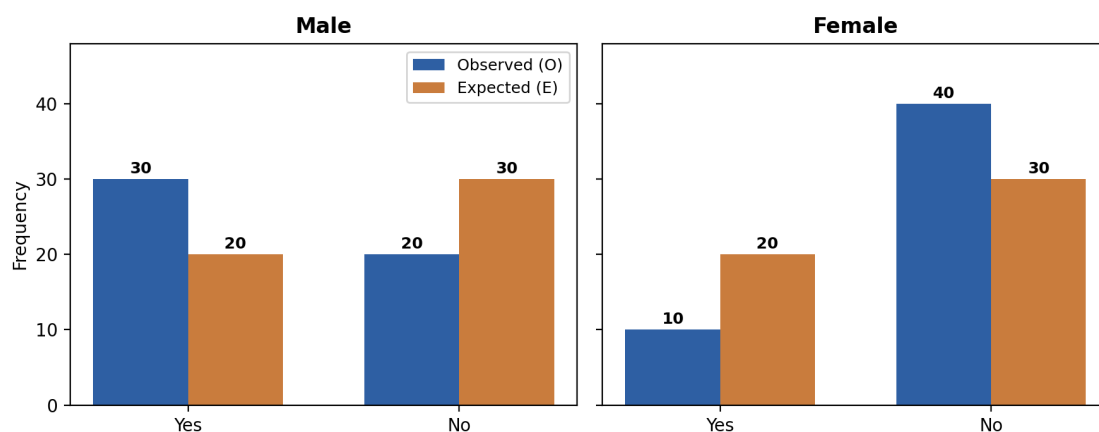


Figure 15.1 — Observed vs Expected frequencies for the Gender × Admission table. Blue = Observed (O), Orange = Expected (E).

Where the blue and orange bars are close, $O \approx E$ (little evidence of association in that cell). Where they diverge strongly — as in every cell of this example — that cell is a candidate for a large contribution to χ^2 . This connects directly back to the formula $\chi^2 = \sum [(O-E)^2/E]$: the larger the visual gap between O and E, the greater that cell's potential contribution to the overall statistic.

15.2 Chi-Square Contribution Visualization

This is explicitly a “Chi-square contribution visualization” — not a generic bar chart. Each bar is one cell's contribution, $(O-E)^2/E$, computed from the same table.

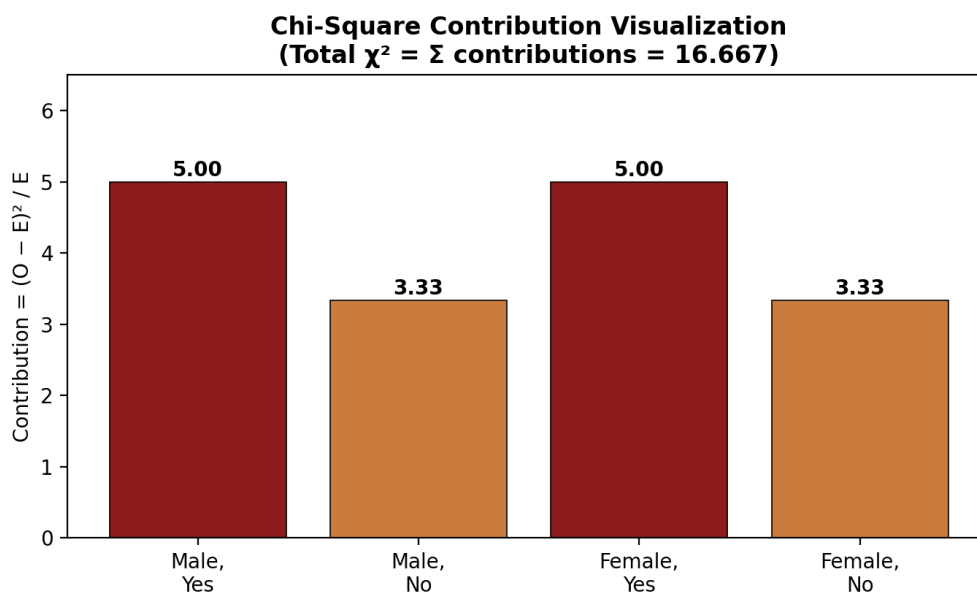
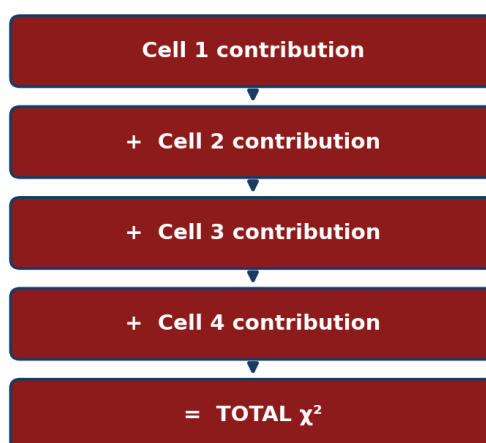


Figure 15.2 — Chi-square contribution of each cell. The four bars sum exactly to the total χ^2 statistic (16.667).

- Small contribution → that cell is relatively close to its expected value.
- Large contribution → that cell contributes more strongly to the overall χ^2 , and is therefore more responsible for any statistically significant result.

Cell Contributions Sum to χ^2



15.3 Residual Heatmap

$$\text{Pearson Residual} \\ R_{ij} = (O_{ij} - E_{ij}) / \sqrt{E_{ij}}$$

- Positive residual ($R > 0$) $\rightarrow O > E$ — this combination occurs more often than independence predicts.
- Negative residual ($R < 0$) $\rightarrow O < E$ — this combination occurs less often than independence predicts.
- Residual close to zero ($R \approx 0$) $\rightarrow O \approx E$ — behaves as independence would predict.
- Large absolute residual $\rightarrow$ strong deviation from independence in that specific cell.

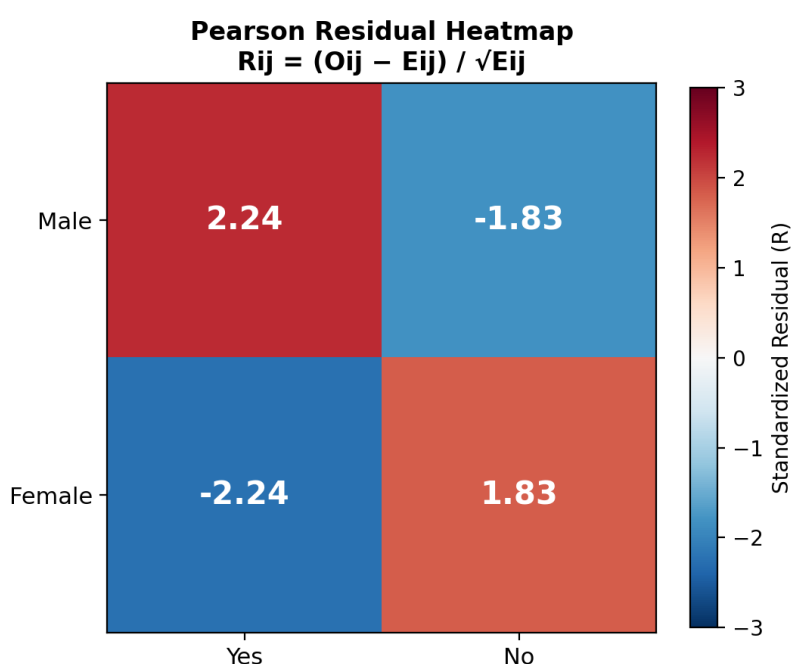


Figure 15.3 — Pearson residual heatmap for the Gender $\times$ Admission table. Red = positive residual ($O > E$), blue = negative residual ($O < E$).

The rows and columns of this heatmap correspond exactly to the categories of the contingency table (Male/Female by row, Yes/No by column). It directly answers: “Which category combinations differ most from what independence predicts?”

The key mathematical connection

$$\chi^2 = \sum R_{ij}^2$$

$$\chi^2 = \sum R_{ij}^2 \text{ — The Key Connection}$$

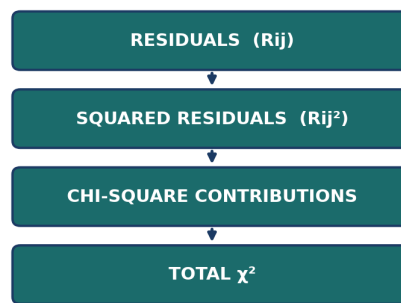


Figure — residuals, once squared, are exactly the chi-square contributions, which sum to the total χ^2 . This is the single most important conceptual link in this whole visualization section.

15.4 Sieve Plot

Sieve plot vs Chi-square test

Sieve Plot = a visualization of the contingency-table structure and the observed-vs-expected relationship.

Chi-square test = a statistical test of evidence against independence.

A sieve plot does NOT itself provide statistical significance — it is a structural/visual diagnostic, not a hypothesis test.

A sieve plot is built from the same contingency table used above. Each row of the plot corresponds to a row category (Male / Female); each column corresponds to a column category (Yes / No). Within a cell, a grid of small squares is drawn so that the number of little squares approximates the expected frequency for that cell, while the observed frequency determines the width/height of the tile itself (tile width is proportional to the column total, tile height is proportional to the observed share within that column). The tile is then shaded — conventionally darker/red for $O > E$ and lighter/blue for $O < E$ — so that deviations from expectation are visible at a glance.

On the Berkeley Admissions sieve plot

The original brief asked to reuse a Berkeley Admissions sieve plot from uploaded material or an image if available. No such file was attached to this request, so no Berkeley-specific sieve plot values are invented here. Section 18 covers the Berkeley Admissions example using the well-known published aggregate figures instead, with sources noted.

**Sieve Plot (Conceptual) — Tile width $\propto$ column total,
tile height $\propto$ observed share, grid density $\propto$ expected count**

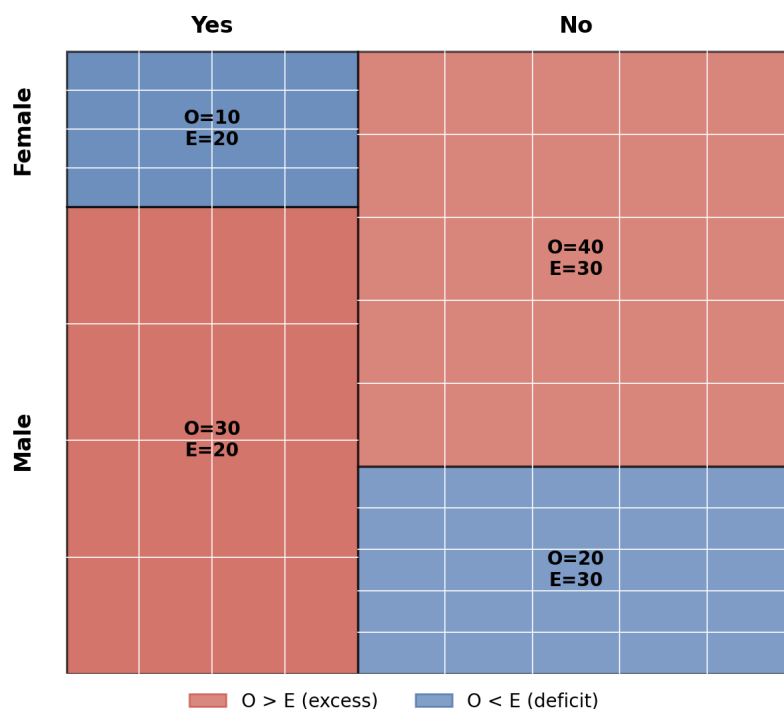


Figure 15.4 — Conceptual sieve plot built from the Gender $\times$ Admission table used throughout this document. Tile area encodes the observed frequency; grid density encodes the expected frequency; color encodes the direction of the residual.

16. SIEVE PLOT VS RESIDUAL HEATMAP

Feature	Sieve Plot	Residual Heatmap
Based on contingency table	Yes	Yes
Related to observed frequencies	Yes	Yes
Related to expected frequencies	Yes	Yes
Shows deviation from independence	Yes	Yes
Shows direction clearly	Limited	Yes
Shows residual magnitude	Limited	Yes
Shows cell-level contribution	Indirect	Strongly
Helps identify unusual cells	Yes	Yes
Easy for beginners	Moderate	High
Useful for Data Analyst interpretation	Yes	Yes

The sieve plot is useful for visually understanding the overall structure of observed vs expected frequencies across the whole table at once. The residual heatmap is especially useful for precisely identifying the direction and magnitude of cell-level deviations. Neither visualization is universally superior — they serve slightly different diagnostic purposes and are often used together.

17. WHAT THE VISUALIZATION DOES NOT DO

IMPORTANT WARNING

A visualization does NOT replace the chi-square test. Visualization $\neq$ Statistical significance.

Correct workflow

CHI-SQUARE TEST → determines evidence of association.

RESIDUAL ANALYSIS → helps identify where deviations occur.

VISUALIZATION → helps communicate those deviations.

18. BERKELEY ADMISSIONS EXAMPLE

On the data used in this section

No uploaded course file or image with specific Berkeley Admissions numbers was attached to this request. Rather than inventing figures, this section uses the well-known published aggregate numbers from the 1973 UC Berkeley graduate admissions dataset (Bickel, Hammel & O'Connell, Science, 1975) — a standard textbook illustration for exactly this topic. If your course material has different or more precise figures, they should replace the ones below.

- Categorical variables: Gender (Male/Female) and Admission Status (Admitted/Rejected).
- Contingency table: rows = Gender, columns = Admission Status, cells = number of applicants of that gender with that outcome, aggregated across all departments.

Aggregate contingency table (approximate, classic dataset)

	Admitted	Rejected	Total
Male	1198	1493	2691
Female	557	1278	1835
Total	1755	2771	4526

What would independence mean here? That the proportion of applicants admitted is the same for men and women once we look at the university as a whole.

Cell	Expected Frequency (approx.)
Male, Admitted	$(2691 \times 1755) / 4526 \approx 1043.6$
Male, Rejected	$(2691 \times 2771) / 4526 \approx 1647.4$
Female, Admitted	$(1835 \times 1755) / 4526 \approx 711.4$
Female, Rejected	$(1835 \times 2771) / 4526 \approx 1123.6$

What does the chi-square test determine here? Running this aggregate table through the chi-square test produces a large χ^2 statistic and a very small p-value — statistically, there is strong evidence of association between gender and admission at the university-wide level.

What additional insight do residuals / deeper analysis provide?

When the same data is broken down department by department, most individual departments show no significant association — some even favor female applicants slightly. Women in this dataset disproportionately applied to more competitive departments (which had lower admit rates for everyone), while men applied more to less competitive departments. This is a textbook case of Simpson's Paradox: an aggregate-level association can weaken or reverse once a hidden variable (here, department choice) is accounted for.

Statistical conclusion vs Visual interpretation

- Statistical conclusion: the aggregate χ^2 test shows a significant association between gender and admission at the whole-university level.
- Visual interpretation: a department-level breakdown (e.g., a grouped bar chart of admit rate by gender, faceted by department) is what actually explains why that aggregate pattern exists — and reveals that it is driven mainly by differing application patterns across departments, not a uniform bias.

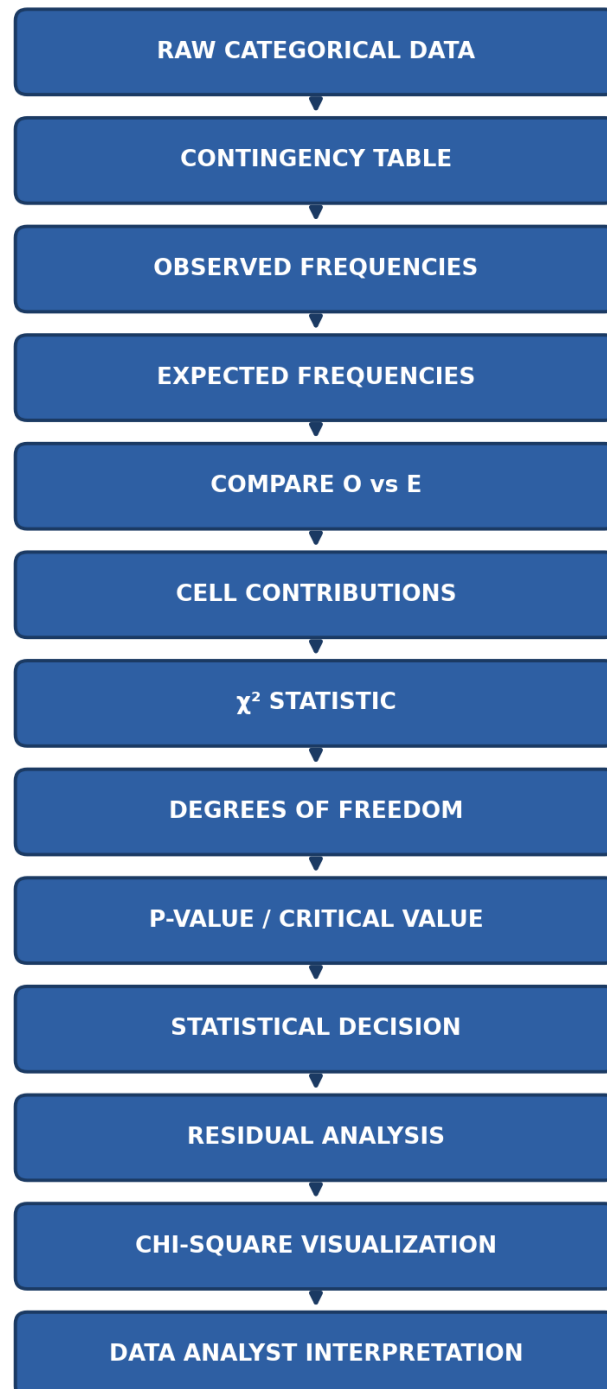
Communicating this to a non-technical stakeholder

“At the university-wide level, admission rates differ by gender in a way that's very unlikely to be due to chance alone. However, when we break the numbers down by department, most departments don't show this pattern — some even show the reverse. The university-wide difference appears to be driven mainly by which departments each group tends to apply to, not by gender bias within departments. We recommend a department-level review before drawing conclusions about bias.”

Association $\neq$ Causation: the aggregate association is statistically real, but it does not mean gender causes rejection — a hidden confounding variable (department) explains most of the pattern. A significant chi-square result only ever shows a statistical relationship in the sample — never a cause.

19. COMPLETE CHI-SQUARE ANALYSIS WORKFLOW

Complete Chi-Square Analysis Workflow



20. PYTHON IMPLEMENTATION

The Python example below uses the exact same Gender × Admission data used throughout this document — no new or unrelated dataset is introduced.

```
import pandas as pd
import numpy as np
from scipy.stats import chi2_contingency
import matplotlib.pyplot as plt

# 1. Contingency table (same Gender x Admission data used throughout)
data = pd.DataFrame({
    "Gender": ["Male"]*50 + ["Female"]*50,
    "Admission": (["Yes"]*30 + ["No"]*20) + (["Yes"]*10 + ["No"]*40)
})
table = pd.crosstab(data["Gender"], data["Admission"])
print(table)
#      No  Yes
# Female  40  10
# Male   20  30

# 2-5. Chi-square statistic, p-value, degrees of freedom, expected frequencies
chi2, p_value, dof, expected = chi2_contingency(table, correction=False)
print("chi2:", chi2, "| p-value:", p_value, "| dof:", dof)
print("expected:\n", expected)

# 6. Pearson residuals
observed = table.values
residuals = (observed - expected) / np.sqrt(expected)

# 7. Chi-square cell contributions
contribution = (observed - expected) ** 2 / expected
print("contribution:\n", contribution, "| sum:", contribution.sum()) # equals chi2

# --- Visualization 1: Observed vs Expected ---
# (grouped bar chart of table.values vs expected, one panel per row category)

# --- Visualization 2: Chi-square cell contribution ---
# (bar chart of contribution.flatten() per cell)

# --- Visualization 3: Residual heatmap ---
fig, ax = plt.subplots()
im = ax.imshow(residuals, cmap="RdBu_r", vmin=-3, vmax=3)
ax.set_xticks(range(len(table.columns))); ax.set_xticklabels(table.columns)
ax.set_yticks(range(len(table.index))); ax.set_yticklabels(table.index)
for i in range(residuals.shape[0]):
    for j in range(residuals.shape[1]):
        ax.text(j, i, f"{residuals[i, j]:.2f}", ha="center", va="center")
fig.colorbar(im, ax=ax, label="Pearson residual")
plt.title("Pearson Residual Heatmap")
plt.show()

# --- Visualization 4: Sieve plot ---
# A basic sieve-style plot can be built by tiling rectangles sized by row/column
# totals (width/height proportional to totals) and shading each tile by the sign
# of its residual (see Section 15.4 for the conceptual construction).
```

Explaining `chi2_contingency()`'s return values

Returned value	Meaning
chi2	The calculated Chi-square statistic (16.667 for this data)

Returned value	Meaning
p-value	Probability of a result this extreme if H_0 were true (≈ 0.0000446)
dof	Degrees of freedom, $(rows-1)(cols-1) = 1$
expected	The expected-frequency table under independence

21. CODE → STATISTICS → VISUALIZATION CONNECTION

Python Output	Statistical Meaning	Visualization
observed	Actual frequencies	Observed table / chart
expected	Frequencies under H_0	Expected table / chart
chi2	Overall deviation	Overall test result
p-value	Evidence against H_0	Decision
residuals	Cell-level deviation	Residual heatmap
contribution	Cell contribution to χ^2	Contribution visualization

Each row of code output maps to one statistical concept, and each statistical concept maps to one specific visualization used in Section 15 — nothing in this chapter is a generic, disconnected chart.

22. DATA ANALYST INTERPRETATION

Avoid

“The variables are dependent.” (too blunt, sounds causal to a non-technical reader)

Prefer

“The Chi-square test provides evidence of an association between X and Y ($\chi^2 = 16.667$, $df = 1$, $p < .001$).”

Then add residual interpretation:

Residual interpretation

“Residual analysis indicates that specific category combinations deviate more strongly from the frequencies expected under independence.”

“These cells contribute more strongly to the overall Chi-square statistic.”

No causal claims should be made at any point in this reporting language.

23. COMMON MISTAKES

Mistake 1

Confusing observed and expected frequencies.

Mistake 2

Using the wrong expected-frequency formula.

Mistake 3

Calculating degrees of freedom incorrectly.

Mistake 4

Reversing the decision rule.

Mistake 5

Thinking the p-value tells us the probability that H_0 is true.

Mistake 6

Treating association as causation.

Mistake 7

Thinking a visualization proves statistical significance.

Mistake 8

Looking only at the total χ^2 and ignoring cell-level deviations.

Mistake 9

Ignoring assumptions.

Mistake 10

Using a generic visualization that is not connected to the chi-square analysis.

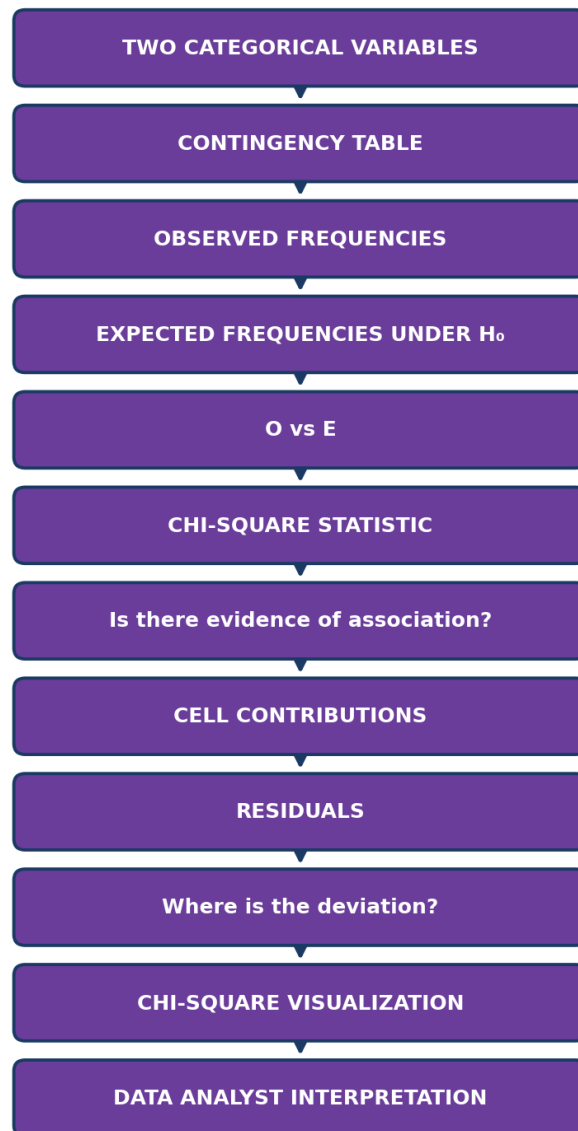
24. PRACTICAL DATA ANALYST EXAMPLE

Complete end-to-end walk-through using the same Gender × Admission Status variables used throughout this document.

Stage	Result
Business Question	Does admission outcome differ by applicant gender?
Categorical Variables	Gender (Male/Female) and Admission Status (Yes/No).
Contingency Table	Male: 30 Yes / 20 No (Total 50). Female: 10 Yes / 40 No (Total 50).
Observed Frequencies	$O_{11}=30, O_{12}=20, O_{21}=10, O_{22}=40$
Expected Frequencies	$E_{11}=20, E_{12}=30, E_{21}=20, E_{22}=30$
χ^2 Calculation	$5.000 + 3.333 + 5.000 + 3.333 = 16.667$
df	$(2-1)(2-1) = 1$
p-value	≈ 0.0000446
Decision	$p\text{-value} < 0.05$ and $< 0.01 \rightarrow \text{Reject } H_0$
Residuals	$R_{11}=+2.236, R_{12}=-1.826, R_{21}=-2.236, R_{22}=+1.826$
Chi-square Visualization	Residual heatmap shows Male over-represented in Yes, Female under-represented in Yes (and the mirrored pattern for No).
Analyst Interpretation	There is statistically significant evidence of an association between gender and admission status in this sample; Male applicants are admitted more often than independence would predict, and Female applicants less often. This is an association, not evidence of a causal mechanism (see Section 13 and the Berkeley example in Section 18 for why a confounding variable must be ruled out before concluding bias).

25. FINAL MENTAL MODEL

Final Mental Model



26. ONE-PAGE QUICK REVISION SHEET

Expected Frequency

$$E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$$

Chi-square Statistic

$$\chi^2 = \sum [(O_{ij} - E_{ij})^2 / E_{ij}]$$

Degrees of Freedom

$$df = (m-1)(n-1)$$

Pearson Residual

$$R_{ij} = (O_{ij} - E_{ij}) / \sqrt{E_{ij}}$$

Cell Contribution

$$\text{Contribution} = (O_{ij} - E_{ij})^2 / E_{ij}$$

Decision Rule

$$p\text{-value} < \alpha \rightarrow \text{Reject } H_0 \quad | \quad p\text{-value} \geq \alpha \rightarrow \text{Do not reject } H_0$$

Decision	Interpretation
Reject H_0	Evidence of association
Do not reject H_0	Insufficient evidence of association

Important

Association $\neq$ Causation.

Visualization Logic

**Observed $\rightarrow$ Expected $\rightarrow$ Difference $\rightarrow$ Contribution $\rightarrow$
Residual $\rightarrow$ Visualization**

27. FINAL 10-POINT SUMMARY

1. The Chi-square test of independence checks whether two categorical variables are statistically associated, using count data organized in a contingency table.
2. Categorical variables are variables whose values are labels/categories, not measured quantities — e.g., Gender, Admission Status.
3. The contingency table cross-tabulates the two variables and provides row totals, column totals, and the grand total needed for every later calculation.
4. Observed frequency (O) is the actual count found in the sample for a given cell.
5. Expected frequency (E) is the count predicted for a cell if the two variables were independent: $E = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$.
6. The Chi-square statistic, $\chi^2 = \sum[(O-E)^2/E]$, summarizes the total mismatch between observed and expected frequencies across the whole table.
7. Degrees of freedom, $df = (m-1)(n-1)$, determine which chi-square distribution is used for the test.
8. The statistical decision compares $\chi^2(\text{calculated})$ to $\chi^2(\text{critical})$, or the p-value to α , to decide whether to reject H_0 (independence).
9. Residual analysis ($R = (O-E)/\sqrt{E}$) identifies exactly which cells are driving a significant result, going beyond the single overall χ^2 number.
10. Chi-square-related visualizations — observed-vs-expected charts, contribution charts, residual heatmaps, and sieve plots — communicate these deviations, but never replace the statistical test itself.

End of document. Formulas, decision rules, and structure follow standard chi-square-test conventions (see the note on Page 1); numbers throughout use the Gender × Admission Status example exactly as given in the brief.